

Mnemonic Tracing: Using Eye Gaze to Search for Visual Memories

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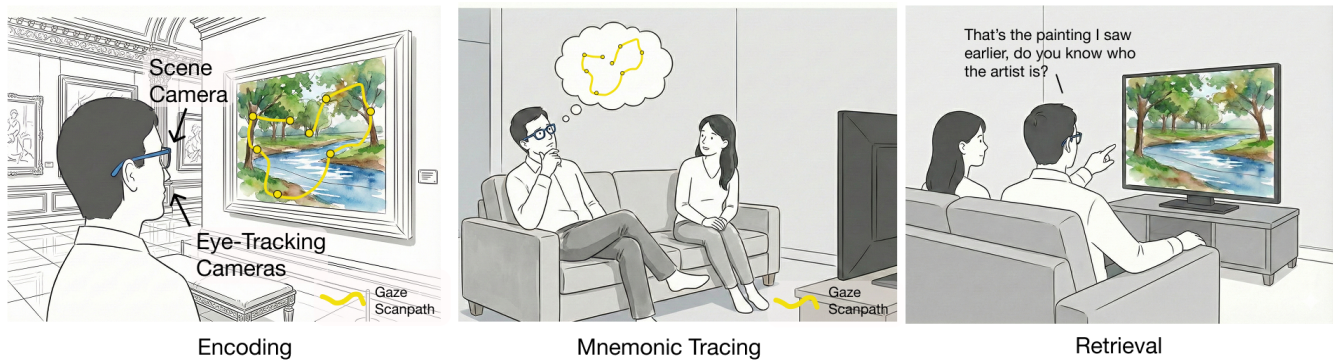


Figure 1: Scenario showing interaction with smartglasses using MNEMONIC TRACING to digitally retrieve an image seen earlier

Abstract

While modern image retrieval systems rely on natural language, some users struggle or are unable to verbalize specific visual memories. Drawing on gaze reinstatement research, we introduce MNEMONIC TRACING: a non-verbal interaction where users intentionally externalize mental images by tracing them onto a blank space using eye gaze. We conducted a pilot study ($N = 11$) to evaluate retrieval of 30 images of people, objects, and locations using eye-tracking glasses. Without training data or prior user training, our system achieved Top-1 accuracy of 30.5% (compared to 3.3% chance level) and a Top-3 accuracy of 51.2%. We show performance is influenced by individual imagery vividness and task engagement, suggesting success is driven by active mental reconstruction. Highlighting its practical potential, participants reported moderately high perceived usefulness ($M = 3.58/5$, $SD = 0.82$) on the Technology Acceptance Model survey. The method offers a private, hands-free, and inclusive alternative for scenarios where a text- or voice-based search is inaccessible.

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CCS Concepts

• **Human-centered computing** → **Interaction techniques**; *Empirical studies in HCI*; *Accessibility technologies*; • **Computing methodologies** → *Computer vision*.

Keywords

Eye tracking, Gaze reinstatement, Gaze-based retrieval, Episodic memory, Implicit interaction

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1 Introduction

We frequently encounter the need to recall visual information from our past, whether we are trying to find a specific photo from a trip, share a discovery with friends, or look up a visual reference for a shopping task. This process is tied to the structure of episodic memory, which is often internally represented as visual images [8]. As we increasingly document our lives through always-on wearable cameras and smartphones, our digital collections become valuable extensions of our own visual memory. While modern systems have made significant strides in retrieval through natural language descriptions [26] and contextual metadata [24], these methods sometimes encounter a "semantic gap". This gap is particularly evident

when a user cannot find the right words to describe a visual memory, or for users with speech or motor impairments who may find traditional query entry methods cumbersome.

Prior work has shown that eye movements reflect underlying cognitive processing [34]. During visual encoding, gaze plays a critical role in binding visual elements and signaling event structure [9, 33]. Psychological research on gaze reinstatement further demonstrates that during recall, people often recreate aspects of these spatiotemporal eye movement patterns while mentally visualizing a scene [22, 29, 38].

While Wang et al. demonstrated that involuntary recall gaze can serve as a retrieval signal [37], they also suggested that the "small effort" of making more extensive, intentional eye movements could significantly improve system accuracy. We lean into this potential by introducing MNEMONIC TRACING, a voluntary interaction where users intentionally trace their mental imagery through eye movements. By transforming gaze from a passive byproduct of recall into an active retrieval tool, we enable a more expressive and controllable interface that bridges the gap between internal mental imagery and digital search. To evaluate this approach, we present a pilot study to examine whether such spatiotemporal gaze patterns can be harnessed as a functional interaction method for visual retrieval. Our contributions include:

- The introduction of MNEMONIC TRACING and a training-free algorithm that ranks images based on spatiotemporal similarity between encoding and voluntary recall gaze.
- A pilot study ($N = 11$) achieving 51.2% Top-3 accuracy, with a qualitative analysis of user experience while MNEMONIC TRACING and its real-world applicability.

2 Related Work

The retrieval of personal visual memories has evolved from manual metadata tagging [15] and associative digital archives [13] to multimodal and LLM-based querying systems [6, 24, 26, 39]. However, when memories are difficult to verbalize, Sketch-Based Image Retrieval (SBIR) demonstrates that spatial representations can serve as effective proxies for memory search [5, 12, 20]. *Mnemonic tracing* extends this paradigm into the ocular domain, enabling users to reconstruct visual memories through gaze without manual input. Research on gaze interaction has traditionally emphasized gaze-as-pointer selection [17, 21, 25], gaze gestures [3, 11], and smooth-pursuit interaction [36]. Beyond explicit control, gaze has been used to infer attention, intent, and cognitive state [2, 4, 7], and more recently as an expressive modality for ocular drawing and spatial manipulation [19, 31]. Systems have also leveraged gaze to recognize objects and user intent [30, 35] and extended to mobile devices [23]. Yet, most gaze-based interfaces rely on externally visible stimuli rather than internally generated mental imagery.

While Wang et al. [37] demonstrated that involuntary recall gaze can support image retrieval, they suggested that performance could improve with intentional eye movements. Building on this insight and recent mnemonic tools such as Eye2Recall [16], we transform gaze reinstatement from an involuntary cognitive byproduct into an intentional interaction technique. By leveraging the link between voluntary mental imagery and spatiotemporal gaze patterns,

MNEMONIC TRACING introduces a hands-free, non-verbal mechanism for navigating personal visual archives.

3 Gaze-Based Retrieval Algorithm

Our retrieval algorithm models memory-guided recall as the reinstatement of coarse spatiotemporal attentional structure. Rather than learning image-specific representations, we directly compare gaze patterns produced during encoding and voluntary recall. Prior work shows that involuntary recall gaze is often sparse, spatially distorted, and difficult to align across trials, motivating deformation-tolerant metrics such as Earth Mover's Distance (EMD) [37]. In contrast, our MNEMONIC TRACING paradigm elicits deliberate and spatially expressive gaze behavior, producing denser and more structured fixation patterns. This allows recall to be modeled as continuous spatial attentional distributions, enabling direct comparison between encoding and retrieval without supervised training.

The algorithm operates on fixation-level features—location, duration, and temporal order—which capture sustained and semantically meaningful visual attention during both encoding and recall. Each eye-tracking trial is represented as a sequence of fixations with location (x_i, y_i) and duration d_i . Fixations are projected onto screen space and converted into a spatial attentional map by accumulating duration-weighted Gaussian responses:

$$H(x, y) = \sum_i d_i \cdot \mathcal{N}((x, y); (x_i, y_i), \sigma^2 I), \quad (1)$$

where σ controls spatial tolerance. The resulting map is normalized such that $\sum_{x,y} H(x, y) = 1$.

To incorporate temporal structure without enforcing strict scan-path alignment, we compute an additional prefix attentional map using only the first K fixations, denoted $H^{(K)}(x, y)$. This representation captures early attentional organization while remaining robust to variability in recall order. Given an encoding trial e and a retrieval trial r , similarity between attentional maps is computed using the Bhattacharyya coefficient:

$$BC(H_e, H_r) = \sum_{x,y} \sqrt{H_e(x, y) H_r(x, y)}, \quad (2)$$

which quantifies overlap between normalized spatial distributions. Final similarity is defined as a weighted combination of prefix-based and full-trial similarity:

$$S(e, r) = \alpha BC(H_e^{(K)}, H_r^{(K)}) + (1 - \alpha) BC(H_e, H_r). \quad (3)$$

This formulation balances spatial and temporal information, and is robust to moderate spatial distortion while supporting subject-specific retrieval. For reference, we include EMD-based comparisons in Appendix D, where the proposed approach consistently outperforms deformation-based matching in the voluntary tracing setting. Algorithm 1 summarizes the retrieval procedure, with additional implementation details provided in Appendix C.

4 Study Design

We conducted a controlled laboratory experiment to evaluate the feasibility of MNEMONIC TRACING for visual retrieval and the gaze-based retrieval algorithm. The study used Pupil Labs Neon eye-tracking glasses [32] to capture eye gaze during encoding and recalling for three categories of images: people, objects, and locations.

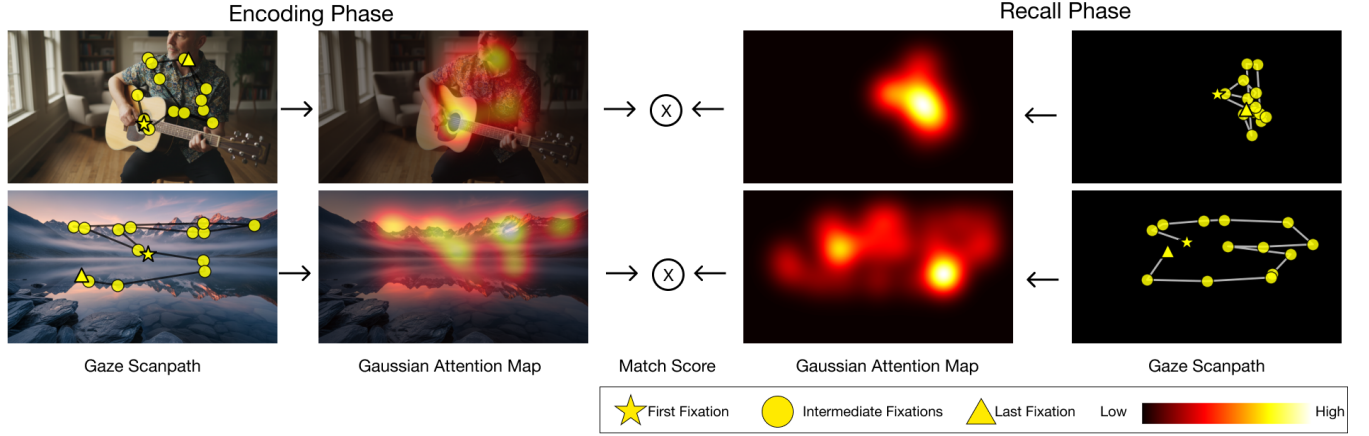


Figure 2: Example of gaze-based retrieval algorithm from study. The scanpaths are from the eye movements during the encoding and retrieval phases of the same image. The gaussian heatmaps are generated which are matched using the Bhattacharyya coefficient.

Algorithm 1 Gaze-Based Retrieval via Gaussian Attention Maps (GAMR)

Require: Encoding trials E_s , Retrieval trials R_s , Gaussian width σ , prefix length K , blend weight α

Ensure: Top- k retrieval performance

- 1: Precompute full and prefix maps with temporal encoding $\{(H_e, H_e^{(K)})\}_{e \in E_s}$
 - 2: **for** each retrieval trial $r \in R_s$ **do**
 - 3: Compute $(H_r, H_r^{(K)})$
 - 4: **for** each encoding trial $e \in E_s$ **do**
 - 5: $S(e, r) \leftarrow \alpha \text{BC}(H_e^{(K)}, H_r^{(K)}) + (1 - \alpha) \text{BC}(H_e, H_r)$
 - 6: **end for**
 - 7: Rank e by $S(e, r)$; record Top- k
 - 8: **end for**
-

Eye-movement events were extracted using the standard processing pipeline. The study sequence of encoding, recall, and experiential phases, lasting approximately 40 minutes, is illustrated in Figure 6.

Participants. We recruited 11 participants (age: $M = 27.27, SD = 10.48$; 5 male, 6 female). Eligible participants were at least 18 years old with normal or corrected-to-normal vision and no eye disorders. Following university ethical approval [Anonymous Approval ID], each received \$20 USD compensation. Individual visual imagery ability was assessed via the Vividness of Visual Imagery Questionnaire (VVIQ) [27], with scores ranging from 53 to 78 ($M = 62.00, SD = 8.15$).

Procedure. Participants were seated 1 m from a 4K display with head position stabilized by a chin rest to minimize head movement. A custom web interface synchronized stimuli with the eye-tracking data. Participants completed six experimental sets, each containing five AI-generated images (examples in Appendix). Detailed instructions provided to participants are included in Appendix, and set order was counterbalanced via Latin square.

During the encoding phase, as shown in Figure 6, participants were shown 3-8 word textual cues followed by the stimuli. They were instructed to memorize the image for 7 seconds while voluntarily tracing the distinctive details using their eye gaze. 7 seconds was determined through empirically to allow for participants to be able to remember the image well in the task. Following a one-minute distractor task, during recall phase, participants mentally projected the image onto a blank screen (‘‘look-at-nothing’’ paradigm [22] to reduce bottom-up distractions) for 7 seconds, tracing details from memory with their eyes. After each trial, participants rated vividness of the recalled image in their mind and confidence that it was the correct image, on 5-point Likert scales. During the experience phase, participants chose one image from the current set to recall a second time. Our algorithm (Section 1) then retrieved and displayed an image based on the trace (took around 3 seconds). Finally, participants completed NASA-TLX [18], Technology Acceptance Model survey [10], and open-ended survey questions.

5 Results & Discussion

5.1 Technical Evaluation

Encoding–retrieval pairs in which participants rated recall vividness or confidence as *Not at all* or *Slightly* were excluded, yielding 295 valid trials out of 330. Applying the gaze-based retrieval algorithm to these trials achieved mean accuracies of 30.5% (Top-1), 51.2% (Top-3), and 60.0% (Top-5), substantially exceeding chance performance (3.3%, 10.0%, and 16.7%, respectively; Table 5). When restricting retrieval to images within the same semantic category, performance remained consistently above chance, with variability across participants (Figure 4A). In a more constrained within-set condition (five images per set), mean Top-1 accuracy increased to 53.2% (chance: 20%). Associations between imagery ability (VVIQ), subjective workload (NASA-TLX), and retrieval accuracy are shown in Figure 4B. During the experience phase, the system correctly retrieved the intended image in 60.6% of 60 trials.

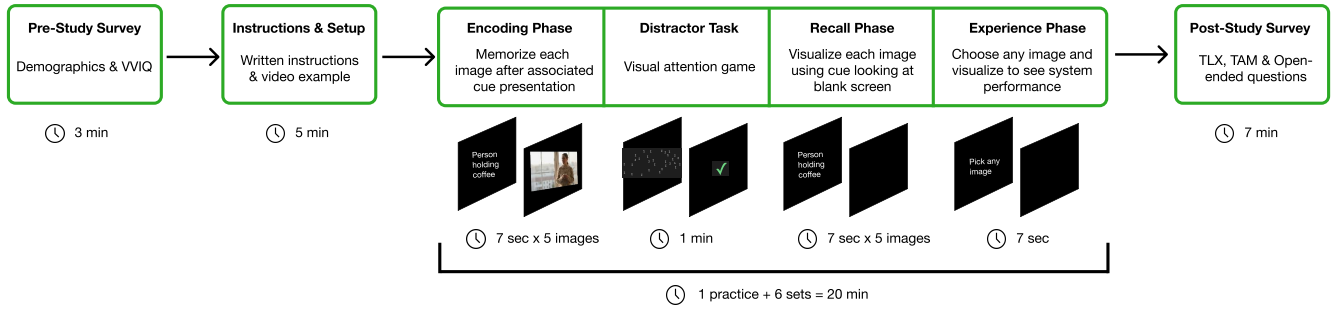


Figure 3: Overview of the study workflow, detailing the sequence and timing of the encoding, distractor, recall, and experience phases.

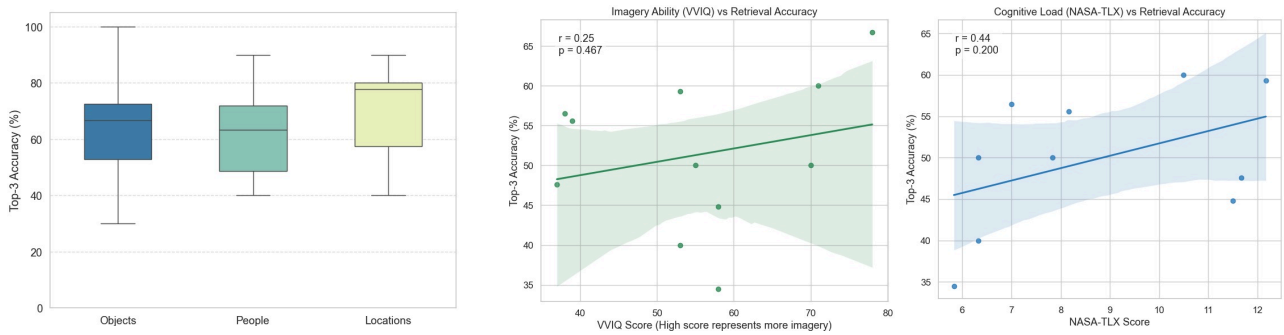


Figure 4: (a) Top-3 retrieval accuracy across image categories (objects, people, and locations). (b) Relationship between participant measures and retrieval performance. Left: Correlation between VVIQ scores and Top-3 retrieval accuracy. Right: Correlation between NASA-TLX scores and Top-3 retrieval accuracy. Shaded regions indicate 95% confidence intervals. Shows higher imagery vividness and increased task engagement associated with improved retrieval performance

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	Overall	Chance
Top-1	36.7	19.0	21.7	27.6	17.2	40.0	33.3	33.3	32.1	26.7	50.0	30.5	3.3
Top-3	50.0	47.6	56.5	44.8	34.5	66.7	59.3	60.0	50.0	40.0	55.6	51.2	10.0
Top-5	63.3	57.1	65.2	48.3	41.4	70.0	70.4	66.7	60.7	56.7	61.1	60.0	16.7

Table 1: Per-participant retrieval accuracy. Chance assumes uniform random selection from 30 images.

5.2 Experience of the MNEMONIC TRACING Process

Participants described MNEMONIC TRACING as an active reconstruction requiring significant focus. Some noted a "tension" (P10) between involuntary movement and intentional control, with P7 and P11 emphasizing the need to trace "object by object" to avoid "zoning out." This engagement is reflected in our quantitative data (Fig. 4), where higher NASA-TLX scores correlate with improved retrieval. Beyond search, P9 viewed the interaction as a "mindful activity" that could encourage users to pay more attention to their surroundings. While sustained effort occasionally caused eyes to wander (P5), the process remains a focused attempt to translate internal imagery into a digital format.

Mapping these images onto a blank canvas involved specific spatial and temporal challenges. Participants sometimes struggled with "exact dimensions" (P7) and "absolute position" (P8), with P3 often "tracing slightly smaller versions" of the original stimuli. Furthermore, P9 observed that visualization occurs "sequentially" and can be "slower" than expected. These accounts align with the positive trend between VVIQ scores and accuracy (Fig. 4), suggesting that vividness aids retrieval. Ultimately, these findings characterize MNEMONIC TRACING as a personalized experience of recalling memory, shaped by an individual's internal clarity and spatial coordination.

5.3 Potential Real-World Applications

Participants reported a moderately high perceived usefulness ($M = 3.58$, $SD = 0.82$) in the TAM subscale, indicating a general openness to adopting gaze-to-recall as a practical retrieval tool. To evaluate the potential utility of MNEMONIC TRACING, we analyzed participant responses regarding potential real-world applications using Braun & Clarke thematic analysis [1], which revealed four primary areas:

- (1) **Memory Aids and Learning:** Participants envisioned the system as a tool for “*painting a picture in my head*” (P5) to aid recall later. It was noted that it could be useful for navigating “*memory palaces that people try to remember a lot of things*” (P3) and for use in “*visual learning settings*” (P7).
- (2) **Image and Video Retrieval:** Users suggested the modality for “*image retrieval in photo files*” (P1) or “*finding a category of pictures related to a memory*” (P11). P8 highlighted a temporal use case: locating a specific moment when one “*did not recall in what order I had taken a specific video or scenery*.”
- (3) **Spatial and Dynamic Navigation:** P4 found the system could be valuable when “*searching for something specific within that [visual] space*.” P9 suggested it is uniquely suited for “*reliving an experience rather than retrieving a fixed stored image*,” specifically by “*trace[ing] the movement of the main foreground object*.”
- (4) **Accessibility and Forensics:** Participants identified applications for users who are “*mute and unable to use your hands*” (P2). Others suggested forensic utility, such as “*remembering faces*” (P10) or recalling an “*attacker’s face*” (P2) when verbal descriptions are insufficient.

6 Limitations and Future Work

While our work demonstrates the feasibility of MNEMONIC TRACING to retrieve previously seen images in a controlled environment, we identify several limitations in the study design and technical implementation.

Firstly, the evaluation utilized a restricted set of static images. In real-world deployments, naturalistic visual stimuli are observed from various angles and may appear structurally similar. Furthermore, the current implementation does not fully account for head movements. In natural settings, gaze is a coordinated effort between the eyes and the head. Future iterations must account for these geometric transformations and body movements to ensure that mental tracing remains robust to unconstrained environments. Secondly, the temporal gap between encoding and retrieval was relatively short. Human memory is subject to decay over time, and it remains to be seen how gaze query fidelity diminishes as the original visual memory fades. Extending this delay to hours or days will help investigate the longitudinal reliability of gaze reinstatement.

Finally, our study population consisted primarily of younger adults. Cognitive visualization and oculomotor control vary across the lifespan, so results may not generalize to older populations. Furthermore, individual variability in microsaccades must be accounted for [14, 28]. Ultimately, this study serves as a step toward a class of non-verbal, memory-augmented interfaces for a naturally-aligned search experience.

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A Pre-Survey Results

Table 2: Participant Demographics and VVIQ Scores

Participant ID	Age	Gender	VVIQ Score
1	56.0	Female	70
2	22.0	Female	37
3	22.0	Female	38
4	21.0	Female	58
5	28.0	Male	58
6	27.0	Male	78
7	21.0	Female	53
8	20.0	Male	71
9	34.0	Male	55
10	21.0	Female	53
11	28.0	Male	39

B Post-Survey Results

C Detailed Algorithmic and Implementation

This appendix provides additional technical details and rationale for the gaze-based retrieval algorithm described in the main paper.

Table 3: Post-Study Survey Results

Measure	M	SD
NASA-TLX (Raw, 1-21)	8.26	2.50
TAM (Overall) (1-5)	3.48	0.82
TAM (Perceived Usefulness) (1-5)	3.58	0.81
TAM (Perceived Ease of Use) (1-5)	3.38	0.97

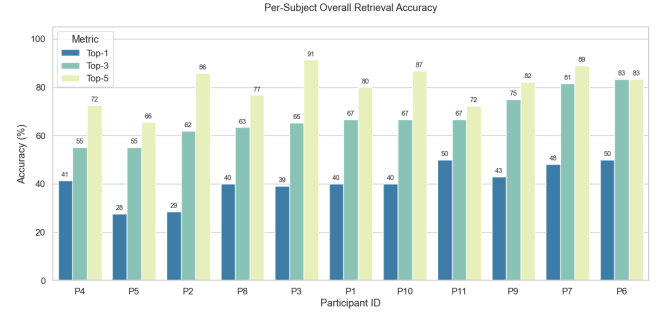


Figure 5: Per-participant Top-1 retrieval accuracy within each category

The goal is to support reproducibility and clarify design choices that were abstracted in the main text for brevity.

Fixation Representation and Preprocessing

Each eye-tracking trial is represented as a temporally ordered sequence of fixations

$$F = \{(x_i, y_i, d_i)\}_{i=1}^N,$$

where (x_i, y_i) denotes the fixation location in screen coordinates and d_i denotes fixation duration. Fixations and saccades were extracted using the Pupil Labs Neon processing pipeline, which applies established velocity- and dispersion-based criteria to raw gaze samples. Only fixation events were used in the retrieval algorithm.

Fixations were sorted by timestamp to preserve temporal order. Trials with missing or invalid gaze samples were excluded during preprocessing. If fewer than K fixations were present in a trial, all available fixations were used when constructing prefix maps.

Attentional Map Construction

Fixations were projected onto a discrete screen-space grid corresponding to the display resolution used during the experiment. Each fixation contributed a duration-weighted impulse at its location, which was subsequently smoothed using an isotropic Gaussian kernel with standard deviation σ :

$$H(x, y) = \sum_{i=1}^N d_i \cdot \mathcal{N}((x, y); (x_i, y_i), \sigma^2 I).$$

Gaussian smoothing serves two purposes: (1) it introduces spatial tolerance to account for small inaccuracies in eye-tracking measurements and recall variability, and (2) it produces a continuous attentional distribution that emphasizes regions of sustained attention rather than isolated fixation points. The resulting map was

Table 4: Participant Feedback and NASA-TLX Scores

Participant ID	NASA-TLX (Raw)	TAM Score	Intentional Eye Movement	Eye Movement Helped
1	7.83	3.12	Strongly Agree	Neutral
2	11.67	1.75	Disagree	Disagree
3	7.00	2.62	Agree	Disagree
4	11.50	3.75	Agree	Strongly Agree
5	5.83	4.25	Strongly Agree	Strongly Agree
6	6.00	4.38	Strongly Agree	Agree
7	12.17	3.50	Strongly Agree	Agree
8	10.50	3.88	Agree	Strongly Agree
9	6.33	3.50	Agree	Disagree
10	6.33	4.50	Agree	Agree
11	8.17	2.62	Strongly Agree	Strongly Agree

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11
Top-1 (%)	36.7	19.0	21.7	27.6	17.2	40.0	33.3	33.3	32.1	26.7	50.0
Top-3 (%)	50.0	47.6	56.5	44.8	34.5	66.7	59.3	60.0	50.0	40.0	55.6
Top-5 (%)	63.3	57.1	65.2	48.3	41.4	70.0	70.4	66.7	60.7	56.7	61.1
Overall (%)				Top-1: 30.5		Top-3: 51.2		Top-5: 60.0			
Chance (%)				Top-1: 3.3		Top-3: 10.0		Top-5: 16.7			

Table 5: Per-participant retrieval accuracy across all categories. Chance performance assumes uniform random selection from a 30-image candidate set.

Subject	Gaussian Attentional Map Retrieval			EMD		
	Top-1	Top-3	Top-5	Top-1	Top-3	Top-5
P1	36.7	50.0	63.3	10.3	20.7	37.9
P2	19.0	47.6	57.1	14.3	42.9	57.1
P3	21.7	56.5	65.2	22.2	44.4	59.3
P4	27.6	44.8	48.3	26.7	40.0	60.0
P5	17.2	34.5	41.4	10.0	20.0	30.0
P6	40.0	66.7	70.0	33.3	50.0	66.7
P7	33.3	59.3	70.4	22.2	44.4	59.3
P8	33.3	60.0	66.7	26.7	50.0	63.3
P9	32.1	50.0	60.7	28.6	50.0	53.6
P10	26.7	40.0	56.7	26.7	50.0	63.3
P11	50.0	55.6	61.1	33.3	50.0	66.7
Mean	30.5	51.2	60.0	24.7	42.3	54.5

Table 6: Comparison of retrieval accuracy (%) between the proposed Gaussian Attentional Map Retrieval (GAMR) method and an Earth Mover’s Distance (EMD), reported per subject and averaged across subjects.

normalized to sum to one, yielding a probability distribution over visual attention.

ordered sequence:

$$H^{(K)}(x, y) = \sum_{i=1}^K d_i \cdot \mathcal{N}((x, y); (x_i, y_i), \sigma^2 I),$$

Prefix-Based Attentional Maps

To capture early attentional structure, which may reflect coarse scene layout and globally salient regions, prefix attentional maps were constructed using only the first K fixations in the temporally

followed by normalization. This design is motivated by prior findings that early fixations often encode high-level spatial structure and may be preferentially reinstated during recall, even when later fixation order varies.

Participant	Top-1 (%)
P1	56.7
P2	42.9
P3	47.8
P4	48.3
P5	41.4
P6	66.7
P7	55.6
P8	56.7
P9	50.0
P10	53.3
P11	66.7
Overall	53.2
Chance	16.7

Table 7: Within-set (6-way) Top-1 retrieval accuracy using the gaze-based retrieval algorithm. Chance performance corresponds to uniform random selection from six items (16.7%).

Similarity Metric

Similarity between encoding and retrieval attentional maps was computed using the Bhattacharyya coefficient:

$$BC(H_e, H_r) = \sum_{x,y} \sqrt{H_e(x,y) H_r(x,y)}.$$

The Bhattacharyya coefficient was chosen because it is symmetric, bounded, and robust to differences in overall fixation density. Unlike divergence-based measures (e.g., KL divergence), it does not require strict distributional assumptions and is less sensitive to sparse regions with near-zero probability. Alternative similarity measures (e.g., cosine similarity, histogram intersection) were explored in pilot experiments but produced less stable performance across participants.

Combined Similarity and Ranking

Final similarity between an encoding trial e and a retrieval trial r was computed as a weighted combination of prefix-based and full-trial similarity:

$$S(e, r) = \alpha BC(H_e^{(K)}, H_r^{(K)}) + (1 - \alpha) BC(H_e, H_r).$$

Encoding trials were ranked by decreasing $S(e, r)$, and retrieval performance was evaluated using Top- k accuracy.

Parameter Selection

The parameters $\sigma = 30$ pixels, $K = 12$, and $\alpha = 0.7$ were selected based on internal validation experiments. These values were chosen to balance spatial robustness, early attentional structure, and overall retrieval stability across participants. Performance was qualitatively stable across a moderate range of parameter values, with these settings yielding the most consistent results.

D Additional Technical Results

E Instructions to Participant

Overview

In this study, you will view a series of images (objects, people, or locations). After viewing, you will recall them while looking at a blank screen. There will be 6 sets of 5 images each. In each set, you will go through three parts: **viewing**, **recalling**, and **experiencing**.

Part 1: Viewing

- A cue (e.g., “A person with a coffee mug in hand”) will tell you what image you will see next.
- Use the cue to remember the image.
- While viewing, please pay attention to the details and **unique** characteristics of the image.
- As you try to remember the image, be **mindful of your eye movements**—where your gaze goes, what you focus on, and how you scan the scene.

After viewing the images within that set, you will move on to the recalling part.

Part 2: Recalling

- You will be shown the same cue again.
- First, take a moment to internally visualize the exact details of the image linked with that cue. Then press **Begin Retrieval**.
- You will see a blank screen for 7 seconds.
- During these 7 seconds, **project the mental image onto the blank screen** by freely moving your eyes while tracing the details you remember.
- Be **very mindful of your eye movements** as you recall—use your gaze intentionally to reconstruct the image. **Use the entire screen to trace the image details with your eyes.**

This will happen sequentially for all 5 images in that set.

Part 3: Experiencing

- After recalling, you will have the opportunity to “experience” the gaze recall method.
- Choose one image from the set that you just saw that you would want to recall. Do not say it out loud.
- When you have it in your mind, press **Begin**.
- During the 7-second blank screen, project the “mental image” onto the screen by freely moving your eyes while tracing the details you remember.
- Be **very mindful of your eye movements**—use your gaze intentionally and use the entire screen to trace the image details.

You only do this part once per set.

Important Notes

- Please keep your head as still as possible using the headrest during the entire study.
- Keep your eyes open during recall and experience phases.

Cue	Image	Cue	Image
A vintage bicycle on a park path		bedroom with a book on bed	
A cutting board with vegetables		Park with an empty bench and tall lamppost	
A person working out in a gym		A person with a ball at his feet	

Figure 6: Example of stimulus used with their cues



Figure 7: A participant using the experimental setup, wearing eye-tracking glasses while viewing images on a desktop display during the study.

F Stimulus Examples

G Study Environment